In the Name of God, the Lord of Knowledge and Wisdom

Environment in the Niyaee Planetary City
(Volume Six)



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"We do not think of the Country; we think of the Earth itself".

"To all the dwellers of the five continents of the Earth,

with a heartfelt wish for a world abundant in justice and serenity for every soul".

Chapter One: The Dawn of Niyaee City; The Golden Hymn of a Bio-Intelligent Civilization

The sun rose slowly over Niyaee City, with a dignity akin to the unfurling wings of the first morning bird. Its warm, golden light slid across the glass-like skin of the towers, transforming their surfaces into gateways of radiance—as if each building were not merely reflecting the sun, but drinking it in, storing it deep within its green, living core. In the early hours of the day, Niyaee City resembled an awakening being: alive, breathing, infused with a gentle pulse flowing through its very fibers.

Ramin stood on the balcony of his new home—a dwelling whose walls were composed of a semi-transparent, semi-living material, responsive to sunlight, its hues shifting with the breath of dawn. He leaned forward, gazing down into the heart of the city, his eyes widening with childlike wonder. What lay beneath him was not a city of streets, congestion, or dust. Instead, a calm, silent layer of pristine surface stretched below, under which the city's veins ran—arteries intertwining underground, pulsing with the synchronized rhythm of hydrogen-powered vehicles, transport robots, and energy pathways.

“Āvā… I can’t believe it,” Ramin whispered. “It’s as if the city is breathing—yet without a sound.”

There was a tremor in his voice, the tremor of one confronted with a magnitude beyond prior imagination.

Āvā, his companion and guide, smiled with a quiet serenity—a smile born of the future, shaped by the calm confidence of one deeply familiar with this new world. She gestured toward the horizon, where green rooftops, transparent pathways, and glass towers unfolded like the colossal petals of a blooming flower.

“Ramin, this is Niyaee City. A city not built upon rigid algorithms, but inspired by the wisdom of a bee colony—a city where order is not imposed, but woven, like threads of honey within a hive.”

In Niyaee City, the governing philosophy of design is biological rather than mechanical; a philosophy rooted not in rigid engineering, but in an organic understanding of life itself. Just as complexity emerges from simple coordination within a bee colony, here too the city has reached extraordinary complexity without chaos or disorder.

Ramin spoke softly, almost to himself.

“A bee colony? So… we’re recreating a natural life form at the scale of a city?”

“Not merely recreating,” Āvā replied. “We are repeating, interpreting, and evolving it. Here, every part of the city has a role—just like every bee. One produces energy, another purifies water, another regulates environmental behavior. All of it flows under the guidance of a central intelligence… much like the queen, who guides the colony without ruling it. Guidance, not domination.”

They entered an underground passageway whose walls absorbed and reflected light like flowing water. Small trains glided past in near silence. Robots designed for cargo transport moved with smooth, synchronized motions along predefined paths. No movement collided with another. This harmony was not enforced; it emerged naturally from real-time decision-making within the system.

What Ramin was witnessing was the true heart of the Niyaee City concept: a city where movement is governed not by traffic lights, fixed rules, or prewritten programs, but by moment-to-moment coordination among its components—much like the adaptive flight

paths of bees within a colony.

“There are no vehicles on the surface?” Ramin asked, still struggling to believe it.

Āvā nodded.

“No. The surface of Niyaee City belongs to people—to feet, bicycles, scooters, and the wind. Heavy and transport vehicles have been moved underground, so that the surface remains a place for living, not merely for passing through.”

Her insight landed gently, yet decisively:

In traditional cities, the ground surface is devoted more to machines than to life. Niyaee City reverses this logic—tools are sent underground, and the surface is returned to humans.

Ramin stood in silence as the soft glow of intelligent signage—changing dynamically based on collective behavior and environmental forecasts—played across his face.

“So people are being controlled?” he asked cautiously. “Their behavior, I mean?”

Āvā laughed—not mockingly, but as one laughs at a

fear humanity has long carried.

“No, Ramin. Control here does not mean the loss of freedom. People behave like a colony, yes—but not through coercion. Through understanding. The system assists people; it does not replace their choices.”

In the philosophy of Niyaee City, human agency is neither erased nor diminished. Every choice made by an individual is instantly registered within a web of environmental impacts, eliciting feedback from the entire system. Humanity is part of the system—just as a bee in a colony is neither a slave nor a ruler, but a vital component of the whole.

When they reached the vertical gardens, Ramin stopped. Walls covered in vibrant flowers rose dozens of meters high, releasing a gentle fragrance that drifted through the air, carrying waves of calm to the senses.

“These walls… they’re alive?” he asked.

“Yes,” Āvā replied. “These vertical gardens are not merely decorative. They are the lungs of the city. They produce oxygen, regulate temperature, store solar energy. In Niyaee City, nothing exists solely to be admired—everything is simultaneously beautiful and

functional.”

At that moment, a small robot passed by, displaying a message on its screen:

“Alert: Energy consumption has increased in the northern sector. Optimization recommended.”

Ramin stared, astonished. A simple message—yet born of a vast network of processing and prediction.

“It’s like the city is speaking to us.”

Āvā murmured, “Niyaee City was not just built—it was born.”

Ramin inhaled slowly.

“This isn’t just a city. It’s an ecosystem.”

“Exactly,” Āvā said. “Here, the environment is not outside the city—it is its essence, its soul. Technology is not the enemy of nature; it is its new language.”

As evening light drifted softly across green surfaces and shadows stretched long, there was no heaviness of dusk—only a living calm, long devoured by noise and congestion in traditional cities.

Ramin had not yet emerged from his initial awe, his mind suspended between reality and an unforeseen future. Āvā took a gentle step forward.

“Look, Ramin. Niyaee City was not constructed—it grew.”

Her voice carried the tone of a quiet interpreter, lifting veils of meaning within the narrative. Ramin paused, watching streams of light flow through the green walls of the buildings—light moving like nectar through the cells of a colossal hive.

With architecture inspired by bio-regulation, Niyaee City is not the product of rigid engineering, but the continuation of an organic logic—as if each building were an organ within a greater body.

“This harmony,” Ramin whispered, “it isn’t natural—yet it doesn’t feel unnatural either. It’s as if an inner order flows through the city.”

Āvā smiled in affirmation.

“The order isn’t natural—but it appears natural. That’s the difference between Niyaee City and traditional cities. There, order is imposed. Here, order is born.”

A small robot passing nearby interrupted their reflection. The display on its chest read:

“Increased energy consumption in the northern sector. Suggested action: optimize movement pathways.”

Ramin repeated softly,

“Optimization… even in how we walk?”

Āvā explained,

“Every small movement is data fed into the colony. In a bee colony, it’s the same—every wingbeat, every flight path, is a fragment of information that fine-tunes the entire system.”

In Niyaee City, the relationship between the citizen and the city is not one of command, but of feedback.

The city learns from its people, and the people learn from the city.

Ramin lifted his head. Transparent, lightweight cable cars drifted silently through layers of air like floating insects. Children playing across the city’s surface had their paths subtly guided by delicate streams of light—light that analyzed their speed and prevented collisions before they could occur.

“Āvā… doesn’t all this control feel frightening?” Ramin asked.

Āvā paused—a pause that revealed thought passing from one layer of understanding to another.

“Control?” she replied calmly. “Yes, from the outside it may seem that way. But this city functions like a bee colony; no movement is imposed upon another. The city merely helps ensure that choices do not produce destructive consequences.”

Here, freedom is not the absence of limits, but harmony with the city’s living context—

freedom aligned with responsibility.

They arrived at the vertical plantings. Tiny droplets of water slid across the transparent glass like tears made of light. Ramin asked,

“Are these only for beauty?”

Āvā smiled.

“No. These are the lungs of the city. Each flower-cell is part of the public ventilation system. Niyaee City has no trees that could become hazards in storms; instead, this plant network simultaneously purifies air, cools the

environment, and stores solar energy."

Ramin placed his fingers against the glass of one vertical garden. He felt a subtle coolness—humid, balanced air flowing gently.

"Āvā… it feels as if the city itself is breathing."

"Yes," she replied softly. "And that breathing depends on us. When energy consumption rises, the city breathes faster. When our environmental behavior is balanced, the city grows calm."

From this interaction emerges a profound inference:

For the first time in the history of urban design, the human being has become part of the city's metabolism; human behavior is now a component of the biology of this colony.

They passed alongside an underground corridor. Ramin watched in awe as small trains and cargo robots moved in flawless order—no noise, no pauses, no congestion.

"So the surface of the city is only for living," he said, "not for cars."

Āvā nodded.

“In traditional cities, land is surrendered to vehicles, and humans must navigate between machines. Here, the surface belongs to human breath.”

Still immersed in the depth of this realization, Ramin asked,

“So… in Niyaee City, is the human at the center?”

Āvā answered gently,

“No. The human is part of the colony. The center is harmony—not the individual.”

The words settled deeply in Ramin’s mind. He suddenly understood: Niyaee City was not built on hierarchy, but on interconnection, coexistence, and interaction.

Sunset light streamed through glass-and-green structures, turning pedestrian paths honey-gold, as if the city itself had become a vast hive, its veins filled with luminous nectar.

Ramin whispered,

“Āvā… I think I’m beginning to understand. This city isn’t just a place to live. It’s a living being.”

Āvā replied with quiet satisfaction,

“Exactly. And you have only seen its first layer.”

The dawn of Niyaee City is not the rise of a new sun, but the emergence of a new form of awareness—

one into which Ramin had only just stepped.

They crossed the central square, where sunset light intertwined across glass surfaces like veins of gold. Ramin inhaled deeply.

“Āvā… even the sounds here are calm.”

The inner implication was clear: he had realized that Niyaee City was designed not only for energetic and spatial harmony, but for mental stillness as well. Sound, movement, and even light reflections were engineered to avoid anxiety and sensory overload.

Āvā smiled.

“Exactly. The city protects its citizens not only environmentally, but psychologically. Every path, every movement, every beam of light is regulated through colony-inspired algorithms.”

Ramin glanced at the bicycle and scooter lanes.

“So even the smallest movement is recorded and analyzed?”

Āvā nodded.

“Yes—and the analysis generates immediate feedback. Just like bees: every individual motion influences the entire colony.”

Niyaee City is a fusion of freedom and accountability—

freedom in choice and action, accountability in impact on the whole system.

They arrived at the city’s educational district, where children learned through lightweight electric bicycles and environmental games.

Ramin asked, astonished,

“Āvā… even children’s games are synchronized with the city’s algorithms?”

“Yes,” she answered. “Every educational activity optimizes energy use, waste production, and movement. Children learn from the beginning that they are part of a greater system.”

Education in Niyaee City is not confined to classrooms; it is continuous, practical, and woven into everyday life.

Every game, every movement, becomes an opportunity for environmental learning and alignment with the colony algorithm.

Excited, Ramin asked,

“So these algorithms can manage larger crises too—like floods or heatwaves?”

“Exactly,” Āvā replied. “The system analyzes real-time data, environmental forecasts, and citizen behavior, and initiates preventive action. Nothing is left to chance.”

As they passed through the district, Ramin noticed that even household waste was collected and processed intelligently. A small robot moved past them, announcing:

“Recyclable materials have increased in the southern district. Collection routes optimized.”

Ramin smiled.

“Like bees… everything is used. Nothing is wasted.”

Āvā confirmed,

“Exactly. Niyaee City optimizes the entire system—even in its smallest details.”

They reached the cable car pathways. Ramin looked toward the sky.

“Is this harmony only for the environment—or does it affect social life as well?”

Āvā smiled.

“Every layer of harmony—from environment to movement to energy—supports psychological and social well-being. People learn they are part of a whole, and that sense of belonging generates cooperation and peace.”

Niyaee City is not merely a technical system, but a **social–environmental ecosystem**, cultivating cooperation, responsibility, and interaction.

As sunset light washed over glass and transparent pathways, Ramin felt he had entered a world where human, technology, and nature were seamlessly attuned.

“Āvā,” he said softly, “life here isn’t just an action. It’s

a living experience.”

She nodded.

“Exactly. And you are only at the beginning. Every movement you make, every choice you take, becomes part of the city’s living ecosystem.”

The dawn of Niyaee City is not merely the beginning of a habitat,

but a rebirth in understanding the relationship between humanity and environment, nature and technology.

Ramin and Āvā entered an underground passage, where the soft sounds of transport robots and flowing electrical energy pulsed gently—like the heartbeat of the city.

Āvā gestured.

“Ramin, this is the city’s energy heart. Every flow is monitored in real time.”

Transparent digital panels along the walls displayed colorful diagrams of solar energy, geothermal power, and wind turbines.

Ramin asked with curiosity,

“So the real-time energy consumption of every district is visible?”

Āvā nodded.

“Precisely. Energy prediction algorithms—based on machine learning and bee-colony-inspired models—can forecast consumption peaks up to fifteen minutes in advance and optimize the energy network accordingly.”

This means that every citizen and every device in **Niyaee City** is a component of the intelligent energy network; every minor act of neglect is corrected by the system, and wasted energy is reduced to a minimum.

Ramin glanced at the transparent water channels flowing beneath the city.

“So these channels aren’t just carrying water, are they?” he asked.

Āvā explained,

“Most of this water is recycled domestic and industrial wastewater, treated through a graywater recirculation system and reused in landscaping, ventilation, and even building cooling systems. This alone reduces

freshwater consumption by up to forty percent.”

Ramin said in amazement,

“So even wastewater is reused for energy and environmental balance…”

Āvā smiled.

“Exactly. Even the excess heat generated from wastewater and building ventilation is transferred via heat exchangers into the geothermal network to produce additional thermal energy. Everything returns to the cycle—just like in a bee colony, where no drop of nectar is ever wasted.”

Niyaee City relies on three primary sources of clean energy:

Small-scale wind turbines distributed across rooftops and elevated corridors,

which analyze urban wind flows and redistribute energy dynamically across districts.

A geothermal system with subsurface heat exchangers,

which harnesses heat from the earth and wastewater to regulate building heating and cooling.

Solar panels and energy storage using advanced lithium-ion batteries and hydrogen fuel systems,

allowing stored solar and hydrogen energy to be returned to the grid during peak demand.

As Ramin studied the energy consumption charts, he asked,

“So if energy use spikes in one district, the system can immediately supply power from another?”

Āvā replied,

“Exactly. And through dynamic load management techniques—Demand Response—the smart grid optimizes consumption, reduces network stress, and prevents sudden blackouts.”

At that moment, Ramin realized that **Niyaee City** was not merely clean, but a truly intelligent organism—one that continuously analyzes and optimizes resources, consumption, and efficiency. Every small movement, every unit of energy use, every human activity affects the entire system.

They reached a vertical garden filled with drought-resistant plants, artificial lighting, and intelligent

ventilation systems.

Ramin asked,

“Are these systems only for beauty, or do they serve environmental functions as well?”

Āvā explained,

“Every plant in this garden is part of the natural ventilation and air purification system. Leaves and intelligent soil absorb carbon dioxide, produce oxygen, and regulate humidity. Smart sensors calculate light and moisture levels based on weather forecasts and energy consumption.”

This means that **Niyaee City** functions as a closed, self-regulating ecosystem: architecture, plants, water, air, and energy all operate in harmony—just like a bee colony, where each bee understands its role and the entire system benefits from individual performance.

Ramin took a deep breath.

“Āvā… this is no longer just a city. It’s a living, intelligent being—and we are part of it.”

Āvā nodded.

“Exactly. And as you continue, you’ll see how these same algorithms can manage even severe environmental crises—heatwaves, pollution, or peak energy demand—without anyone even noticing.”

As they crossed the central square, an environmental alert appeared on a transparent display:

“Humidity levels in the eastern district have decreased. Risk of vegetation stress detected. Recommendation: activate artificial humidification and water recirculation systems.”

Ramin asked in surprise,

“Does this mean the city monitors even the smallest humidity changes in real time?”

Āvā nodded.

“Precisely. City-wide IoT sensors—including soil moisture, temperature, solar radiation, and wind flow sensors—transmit data to the central network. Machine learning algorithms analyze this data and determine immediate responses.”

This system is not merely reactive, but predictive and preventive; every environmental fluctuation is

identified and managed before it becomes a real threat.

Ramin watched the transparent water channels continue their calm flow.

“So this water returns to the cycle again?”

Āvā smiled.

“Yes. The graywater and rain-harvesting systems maintain soil and air humidity within a ±3% accuracy range. Even excess heat or intense solar radiation is predicted by thermal sensors, allowing ventilation systems to adjust proactively.”

Ramin said softly,

“So the city can keep vegetation alive—even under harsh natural conditions.”

Āvā continued,

“Exactly. And during critical moments, the energy load management algorithm intervenes. If peak energy consumption rises, geothermal, solar, and wind networks activate in balance to ensure uninterrupted humidity control and ventilation.”

Soil moisture sensors: real-time measurement with data

transmission every 10 seconds

Smart ventilation control: temperature and humidity regulation using advanced PID algorithms

Graywater and rain recirculation systems: up to 40% reduction in freshwater consumption

Integrated energy network: simultaneous use of geothermal, solar, and wind energy to support environmental systems

Ramin looked at the vertical gardens and green pathways and said,

“This goes beyond managing water and energy—it’s a form of living harmony.”

Āvā replied,

“Exactly. And citizen behavior is also part of this system. Algorithms continuously analyze household energy use, water consumption, and environmental activities, offering intelligent recommendations.”

Ramin murmured,

“So if someone is careless, the system corrects it?”

Āvā nodded.

“Yes—without coercion. Pathways, ventilation, and even artificial lighting schedules adjust to maintain city-wide balance. Just like in a bee colony, where every bee is part of the whole.”

This moment marks the peak of the fusion between art and science in **Niyaee City**—a place where environmental systems, energy networks, human behavior, and architecture have all merged into a living, self-regulating organism. Every minor change produces an immediate, optimized response.

Ramin glanced at another digital display showing air quality and energy consumption charts. Green, yellow, and red lines shifted rapidly, yet never crossed the boundaries of control.

“So even crises are managed without serious disruption,” he said.

Āvā replied,

“Exactly. Heatwaves, drops in humidity, surges in energy consumption, even excessive crowding—everything is analyzed and managed in real time. This is the key advantage of the **Niyaee Planetary City**.”

Ramin thought to himself:

“If all cities could follow such a model, no environmental crisis would ever be unmanageable…”

The evening light fell across glass surfaces and urban greenery, and Ramin felt as though every ray of light, every leaf, and every stream of water was speaking to him: **Niyaee City is alive, intelligent, and attentive to even the smallest detail.**

Ramin and Āvā arrived at the central district, where the city surface glowed softly under the late-afternoon sun. Children’s laughter blended with the gentle movement of air. Yet what stood out most was the precise harmony between human movement and the intelligent system.

“Āvā, how is it possible for all this to exist without congestion or chaos?” Ramin asked.

Āvā smiled.

“Ramin, look closely. Every citizen is part of the colony. Colony-inspired algorithms continuously adjust movement paths and activities. Children, adults, and the elderly all move naturally in harmony.”

Here, the concept of the city as a **social–**

environmental ecosystem becomes clear. Citizens are not merely users, but active and responsible participants in maintaining and optimizing the system. Every individual decision—small or large—affects the entire city.

Ramin watched a group of teenagers collecting waste and sorting recyclables.

“Is this educational as well?” he asked.

Āvā replied,

“Yes. Every daily activity—from waste collection to garden irrigation—is logged and analyzed by the system. Even games and recreational activities are part of environmental education.”

Each piece of waste carries a smart identifier, allowing sensors to detect its type, volume, and composition.

The city’s algorithms optimize collection and recycling routes, reducing energy consumption by up to 30% while maximizing material recovery.

Voluntary and educational activities feed behavioral data into the intelligent system, generating real-time feedback.

Smiling, Ramin said,

“So people truly feel that they’re part of the entire system.”

Āvā nodded.

“Exactly. A sense of belonging and responsibility—something traditional cities lack. When citizens know their actions directly affect everyone’s quality of life, participation becomes natural and spontaneous.”

They reached the bicycle and scooter pathways, where small robots and surface sensors predicted movement patterns and coordinated routes based on speed, density, and environmental conditions.

“So every route and timing is scientifically calibrated?” Ramin asked.

“Precisely,” Āvā answered.

“AI algorithms analyze real-time data to balance population density, air quality, and energy consumption at every moment. No congestion, no unnecessary consumption, no environmental strain.”

This harmony is a synthesis of social psychology, environmental management, and intelligent

technology. **Niyaee City** is not merely a physical space, but a dynamic, self-regulating social system—where citizens and artificial intelligence collaborate rather than compete.

Looking at the vertical gardens and green pathways, Ramin said softly,

“Āvā, this place truly feels like collective, synchronized life… as if everyone is breathing together.”

Āvā smiled.

“Exactly. Just as every bee is a living cell of the colony, every citizen of **Niyaee City** is a living cell of the ecosystem. Education, participation, and algorithms work together to sustain balance and well-being.”

Social interaction in **Niyaee City** is neither imposed from above nor purely cultural; it is the result of intelligent design that transforms individual behavior into positive feedback for the entire system.

After a brief pause, Ramin said,

“Āvā… now I understand. This city’s superiority over traditional cities isn’t just technological—it lies in

harmony with nature, reduced traffic, optimized resource use, and a shared sense of responsibility."

Āvā nodded.

"Exactly. When all these elements come together, **Niyaee City** becomes not only clean and intelligent, but a truly living, sustainable experience for all its citizens."

As the evening light washed over transparent pathways and greenery, Ramin felt that every movement, every decision, and every breath of the citizens was part of a living ecosystem—harmonized through algorithms and nature alike.

Ramin and Āvā moved toward a hill overlooking the city. Ramin gazed at the view: transparent pathways, vertical gardens, lightweight scooters and bicycles, and the silent, orderly underground network. His breath caught.

"Āvā… everything here is different. But what exactly makes this city superior to traditional ones?"

Āvā replied calmly,

"Ramin, the comparison can be made across several

dimensions: energy, environment, transportation, citizen behavior, and social sustainability.”

1. Energy and Environment

In traditional cities, fossil fuel consumption leads to pollution and rising urban temperatures. In **Niyaee City**:

– An integrated energy network combining solar, wind, hydrogen, and geothermal sources

– Real-time energy management using machine learning and colony-inspired algorithms

– Waste and water recirculation reducing resource loss by over 40%

This results in near-zero pollution and maximum energy efficiency—something unattainable in conventional cities.

2. Transportation

Traditional cities suffer from traffic congestion, emissions, and accidents.

In **Niyaee City**:

– Heavy vehicles operate underground

– Electric scooters and bicycles are managed by real-time routing algorithms

– Intelligent crowd-density control eliminates the need for manual restrictions

This system not only protects the environment but also provides a calm, psychologically healthy urban experience.

3. Education and Citizen Behavior

In traditional cities, environmental education is limited and passive.

In **Niyaee City**:

– Every daily activity is educational—even children's games

– Algorithms provide instant feedback on citizen behavior

– A sense of responsibility and participation is cultivated from early childhood

Citizens are not merely consumers, but active

contributors to system optimization—something largely absent in traditional cities.

4. Crisis Management and Sustainability

Traditional cities often respond slowly and chaotically to floods, heatwaves, or energy surges.

In **Niyaee City**:

– IoT sensors and intelligent networks predict crises and respond instantly

– Energy and humidity are optimized even under extreme conditions

– Humans and machines operate in complete coordination, without disruption

Niyaee City is both predictive and preventive—like a living organism, continuously balancing environmental pressures and resources.

Looking down at the city, Ramin said,

“Āvā… so the true advantage here isn’t just technology. It’s the harmony between humans, nature, and artificial intelligence that makes this city alive and sustainable.”

Āvā smiled and affirmed,

“Exactly. This city is a living ecosystem—algorithms, human behavior, architecture, green space, and energy all in perfect alignment. Even the smallest movement leaves a trace and creates an impact.”

The dawn of **Niyaee City** is a symbol of the coexistence of humanity and nature within an intelligent technological framework. This city demonstrates how, through precise design and artificial intelligence, it is possible to create an urban environment that is clean, sustainable, psychologically harmonious, and self-regulating—something that remains unattainable in traditional cities, even with modern technology.

Ramin took a deep breath and said,

“If all the cities of the world could follow such a model… pollution, traffic, and environmental stress would no longer exist.”

Āvā nodded in agreement.

“And that,” she said, “is the fundamental goal of the **Niyaee Planetary City**: a life that is harmonious, intelligent, and sustainable—where every individual,

every movement, and every decision forms part of a living, resilient system."

The final rays of sunset fell upon the transparent pathways and urban greenery, and Ramin felt that every particle of light, every leaf, and every flow of water was alive in perfect harmony—guided simultaneously by algorithms and nature.

End of the Day and Reflection

Niyaee City stands as a practical embodiment of intelligent urban systems inspired by swarm-based algorithms and machine learning. In this city, every citizen, every device, and every unit of energy or water consumption is considered an integral component of a dynamic, self-regulating ecosystem.

All elements of the city behave like cells within a bee colony: environmental data, resource consumption, and population density are continuously collected, analyzed, and optimized in real time through colony-based algorithms.

An intelligent resource management network governs energy, water, air circulation, lighting, and material recycling. Using real-time data and predictive models,

equilibrium is achieved without the need for direct human intervention.

This design transforms the concept of dynamic environmental sustainability from theory into operational reality. Unlike traditional cities—where systems are reactive and often passive—**Niyaee City** is predictive and self-regulating.

The energy system of **Niyaee City** integrates geothermal, solar, wind, and hydrogen-based sources. Algorithms continuously analyze energy consumption across neighborhoods, buildings, and lightweight vehicles.

Load balancing algorithms determine which energy source should supply each consumer, ensuring that the entire network remains within optimal sustainability thresholds.

Machine learning models predict consumption peaks and activate alternative energy sources in advance.

Citizens receive intelligent, non-intrusive messages that guide energy optimization without creating a sense of restriction or coercion.

As a result, energy loss is reduced to near zero and

overall network efficiency reaches its maximum. In contrast, traditional cities typically manage energy reactively, responding to crises after consumption peaks generate stress and pollution.

Through IoT sensors and adaptive algorithms, **Niyaee City** continuously monitors water quality and environmental humidity.

Soil and atmospheric humidity sensors transmit data every ten seconds, enabling precise analysis of vegetation and ground conditions.

A water recirculation system reuses greywater and harvested rainwater for irrigation and environmental cooling.

Artificial ventilation and humidity control are maintained using PID controllers and predictive models, keeping humidity within ±3% and preventing vegetation stress.

This precise regulation ensures long-term ecosystem stability while reducing freshwater consumption by up to 40%. By aligning these processes with clean energy systems, additional resource demands are minimized and localized climate impacts are significantly reduced.

In **Niyaee City**, waste is collected, categorized, and transformed into energy or recyclable materials through intelligent systems.

Each waste unit carries a smart identifier that allows sensors to determine its type, volume, and composition.

Algorithms optimize collection and processing routes to achieve maximum efficiency with minimal energy use.

Even subtle changes in citizen behavior are recorded and translated into system feedback.

Much like a bee colony, no resource is wasted; every component contributes to optimizing the entire ecosystem.

The surface and subsurface transportation networks of **Niyaee City** are managed through predictive density algorithms that synchronize human movement and lightweight vehicles.

Electric bicycles, scooters, and cable transport systems align their energy consumption with the intelligent grid.

Heavy vehicles and freight transport are confined to

underground corridors, preserving a clean and congestion-free surface.

Real-time population density algorithms optimize routes and travel times, preventing unnecessary strain on air quality or resources.

This coordination reduces air pollution, lowers energy consumption, and enhances citizen safety—an outcome that even modern cities struggle to achieve without intensive human intervention.

Through intelligent feedback systems and educational environmental games, **Niyaee City** continuously refines citizen behavior.

Every activity—from children's play to household consumption—is recorded and analyzed.

Algorithms provide immediate feedback and optimization suggestions.

Social participation and environmental responsibility are cultivated across all age groups.

This system seamlessly integrates environmental education into daily life, transforming citizens into active agents of sustainability.

Niyaee City is capable of anticipating crises such as heatwaves, humidity loss, surges in energy demand, or excessive population density.

Predictive models operate using both historical and real-time data.

Algorithms dynamically optimize energy distribution, ventilation, humidity control, and human movement.

Responses are executed simultaneously and autonomously, without requiring human intervention.

Unlike traditional cities—where crises often trigger disorder and resource strain—**Niyaee City** functions as a preventive, self-regulating environmental response system.

Ultimately, **Niyaee City** represents a functional model of an intelligent, sustainable, and self-regulating urban ecosystem in which:

1. Energy resources are optimized through predictive and swarm-based algorithms.
2. Water and environmental humidity are continuously monitored and recirculated.
3. Waste is converted into energy and reusable

materials.

4. Urban and human transportation systems are coordinated, clean, and safe.
5. Citizen behavior is analyzed, optimized, and reinforced through continuous environmental education.
6. Environmental crises are anticipated and rapidly managed.

This city not only preserves the environment, but—through intelligent algorithms and human collaboration—offers a scalable and replicable model for the future of urban life. Artificial intelligence, algorithmic intelligence, and citizen participation converge to form a living, dynamic ecosystem: a city in which all components are interconnected, interdependent, and cooperative.

Chapter Two

Artificial Intelligence and the Bee Colony Paradigm in Service of the Environment

The morning mist danced gently through the air of **Niyaee City**, mingling with delicate droplets of dew that settled upon leaves and the cool metallic surfaces of urban structures, refracting the light into countless shimmering hues.

Leila and Saman stood within the city's intelligent management center—a space where hexagonal digital walls and displays, reminiscent of a colossal honeycomb, silently recorded the rhythms of urban life with flawless order and harmony. Each hexagon seemed like a window into a thousand unfolding stories: the smooth movement of intelligent vehicles, the silent and graceful flight of municipal service drones, and the quiet pulse of a city fully aware of itself.

A soft hum emanated from the intelligent management systems, echoing through the space like the gentle whispers of a living organism. Leila extended her hand toward one of the panels; green and golden light flowed past her fingertips as invisible data transformed into

vivid, dynamic visualizations across the displays. Saman observed with a contemplative gaze and a calm smile, watching a city where every pathway and every beam of light resonated with the heartbeat and breath of its people.

The atmosphere was filled with tranquility and hope, as if time itself had paused—just long enough to witness the beauty and harmony of a city where humanity and nature had become one.

“Look, Saman,” Leila said softly. “Every movement of people, every light turning on or off, every bicycle in motion—each has a direct impact on the entire system.”

She continued, “Just like a bee colony, every part of the city—from energy and water to transportation—is part of a living whole. Colony-based algorithms analyze data in real time and distribute resources optimally.”

Scientific Insight:

Bee Colony Optimization (BCO) algorithms in urban management enable multivariable coordination between resources and human activity. Unlike traditional cities, where systems are reactive, these

algorithms are predictive and self-regulating. Every citizen and every subsystem functions as a cell within a dynamic ecosystem.

Saman turned his attention to a holographic display illustrating real-time energy flows across the city. Green, yellow, and red lines traced the movement of power, while algorithms autonomously optimized energy pathways.

“What happens,” he asked, “if energy consumption in a neighborhood exceeds its threshold?”

Leila smiled. “The system immediately rebalances distribution. Light vehicles shift to alternative sources—wind, geothermal, or hydrogen. Building ventilation and lighting are adjusted, and consumption is optimized.”

Predictive energy algorithms continuously anticipate fluctuations in demand. BCO algorithms maintain equilibrium across energy resources, minimizing waste.

Intelligent feedback provided to citizens encourages optimal consumption without coercion, transforming efficiency into a collective learning process.

As a result, the entire urban ecosystem functions as a single dynamic organism—unlike traditional cities, where even minor disruptions can trigger pollution or systemic crises.

Inside the vertical greenhouses, Leila explained:

“These plants are not merely decorative. They play a crucial role in air purification, oxygen production, and urban humidity regulation.”

Sensors monitoring humidity and temperature transmit data to artificial intelligence systems, which optimize water usage, lighting, and ventilation. Storm-resistant vegetation not only produces oxygen but also reduces the energy demand of cooling and ventilation systems.

Predictive algorithms compensate for environmental changes instantaneously—much like a bee colony that rapidly adapts to resource shortages.

Saman looked at Leila and asked, “So even household waste is converted into energy?”

“Exactly,” she replied. “Each waste unit carries a smart identifier. Sensors analyze its type, volume, and composition. Algorithms optimize collection and processing routes, and even the smallest behavioral

changes by citizens are recorded and fed back into the system."

Waste is transformed into energy, fertilizer, or recyclable materials. BCO algorithms optimize waste collection paths and reduce energy consumption in processing facilities.

The feedback system empowers citizens to actively participate in resource conservation and energy efficiency.

Along the bicycle paths and cable transit routes, Leila continued:

"All lightweight transportation systems are synchronized through artificial intelligence. There are no heavy vehicles on the surface of the city."

Population density and movement-flow algorithms continuously adjust routes to prevent congestion and pollution. Electric light vehicles synchronize their energy usage with the smart grid, ensuring maximum efficiency and minimal environmental impact.

The system behaves like a living organism, where each cell contributes to maintaining balance across the entire ecosystem.

Informational Architecture and Sensor Networks

Leila and Saman entered one of **Niyaee City's** primary corridors, where a vast network of IoT sensors, wireless 5G communication, and cloud-based data processing systems monitored every environmental and behavioral detail.

"Notice this, Saman," Leila said. "All data is transmitted to the central artificial intelligence in real time."

"So everything is managed instantaneously?" he asked.

"Exactly. Bee Colony Optimization algorithms regulate energy, water, ventilation, and even social behavior. Every consumer unit or urban sensor functions like a bee—sending feedback and receiving immediate responses. This is what makes the system self-regulating and minimizes resource and energy waste."

The sensor network continuously collects environmental data—temperature, humidity, air quality, energy consumption, and traffic flow.

BCO algorithms process this data in a distributed manner, optimizing energy and resource pathways.

Unlike modern cities, **Niyaee City** operates as a predictive and self-regulating system, preventing sudden stress on infrastructure.

Saman studied the energy diagrams again—green, yellow, and red lines representing solar, wind, geothermal, and hydrogen power.

“What if energy demand peaks in one neighborhood?” he asked.

Leila explained, “BCO algorithms distribute alternative resources dynamically. Ventilation, lighting, and light-vehicle movement are adjusted in real time so total city consumption remains within optimal limits.”

Each consumer unit transmits real-time data. Algorithms perform cumulative optimization using objective functions that minimize energy consumption and loss.

This rapid response allows the system to absorb demand peaks and prevent overload—unlike the reactive and inefficient grids of modern cities.

Passing through transparent walkways, they arrived at the vertical gardens. Each plant was equipped with sensors measuring humidity, temperature, light

intensity, and air quality, transmitting data to the central artificial intelligence.

“These plants,” Leila said, “don’t just beautify the city—they regulate its microclimate.”

“So water consumption decreases as well?” Saman asked.

“Precisely. The algorithms optimize irrigation, ventilation, and artificial lighting pathways. Even the energy buildings consume for cooling is significantly reduced.”

Sensor data is continuously transmitted to reinforcement learning algorithms and crisis-prediction models. Through microclimate modeling and simulations of plant evapotranspiration and water absorption, the algorithms dynamically adjust irrigation and ventilation pathways in real time.

This level of environmental management reduces energy and water consumption by **30–40 percent** compared to traditional cities.

The waste management center of **Niyaee City** was equipped with autonomous robots and intelligent channels.

“Leila,” Saman asked, “how do the algorithms decide where each type of waste should go?”

Leila explained, “Sensors identify the type, volume, and chemical composition of every waste unit. The algorithms then determine whether it should be recycled or converted into energy. Immediate feedback is even sent to citizens to help refine and improve their environmental behavior.”

Colony-based algorithms optimize waste flows and establish a closed-loop energy cycle. This process reduces greenhouse gas emissions, increases energy efficiency, and transforms environmental education into a practical, lived experience.

Unlike modern cities, resource loss is minimized, and the system operates as a self-regulating and adaptive organism.

Leila and Saman stood inside the energy center of **Niyaee City**, observing the flowing lines of solar, wind, geothermal, and hydrogen power.

“Saman,” Leila said, “this system is designed so that energy consumption never exceeds the network’s capacity.”

“How is that possible?” Saman asked.

“Bee Colony Optimization algorithms monitor every consumption unit and distribute energy resources optimally. Any change in energy demand is corrected instantly.”

In modern cities, peak consumption strains the grid and increases reliance on fossil fuel generation. In **Niyaee City**, real-time and self-regulating energy distribution minimizes waste and reduces CO_2 emissions to near zero. Even during critical conditions, intelligent algorithms seamlessly redirect alternative energy sources—wind, solar, geothermal, and hydrogen—to maintain stability.

Across **Niyaee City**, no heavy vehicles were visible. Bicycle lanes, electric scooters, and cable transport systems operated entirely on clean energy.

“So traffic and pollution here are almost nonexistent?” Saman asked.

“Exactly,” Leila replied. “Light electric vehicles coordinate their routes and speeds through real-time algorithms. Beneath the city surface, freight and service vehicles operate discreetly, keeping the urban

environment clean and safe."

The reduction of traffic and air pollution significantly improves citizens' physical and psychological well-being. Bee colony algorithms synchronize vehicle movement, reducing accident risks and traffic pressure.

Modern cities, dominated by heavy vehicles, suffer from higher noise pollution and greenhouse gas emissions.

Leila guided Saman toward the vertical gardens and resilient green coverings.

"These plants regulate the city's microclimate and reduce water consumption," she said.

"So even water use and building ventilation are optimized?" Saman asked.

"Precisely. The algorithms adjust irrigation and ventilation pathways moment by moment, significantly lowering energy consumption."

In modern cities, water loss is high and microclimate management is incomplete. In **Niyaee City**, smart systems driven by IoT sensor data predict and regulate humidity and temperature changes with precision.

Reduced water and energy consumption preserves natural resources and strengthens environmental sustainability.

The city's recycling center was equipped with intelligent robots capable of identifying every type of waste and determining whether it should be recycled or converted into energy.

"How does this system reduce waste so effectively?" Saman asked.

"Every citizen and every sensor cell sends real-time data," Leila explained. "The algorithms optimize waste flows, and immediate feedback is provided to citizens."

In modern cities, waste is often buried or incinerated, resulting in massive resource loss. In **Niyaee City**, a closed-loop waste-to-energy cycle reduces greenhouse gas emissions and conserves resources.

The algorithms function like a bee colony: every unit plays its role, and the entire system achieves high efficiency.

Saman glanced at the crisis-monitoring display, where temperature, humidity, and air quality across all districts were tracked in real time.

“What happens if the temperature in a neighborhood rises?” he asked.

“The system automatically adjusts ventilation, irrigation, and energy resources,” Leila replied. “Light vehicles switch to optimal energy sources as well.”

Environmental crisis prediction algorithms, much like bee colonies responding to threats, react rapidly using real-time data. In modern cities, responses are often delayed and reactive, placing immense strain on infrastructure and resources.

Niyaee City is predictive and self-regulating, managing crises without disrupting daily life.

In the cultural district, children and teenagers engaged with environmental games enhanced by intelligent feedback systems.

“What impact do these games have?” Saman asked.

Leila smiled. “From childhood, citizens learn how to optimize their use of energy, water, and food. The algorithms analyze their behavior and gently guide it toward sustainability.”

Hands-on education and real-time feedback align

individual behavior with the functioning of the urban ecosystem. The algorithms simultaneously assume the roles of educator, coordinator, and predictor, guiding both learning and action in a seamless loop.

In most modern cities, environmental education remains limited, and its impact on everyday behavior is minimal.

The surface of **Niyaee City**, free from heavy vehicles and defined by green corridors and clean air, offers a fundamentally different experience of urban life.

“I feel that life here is truly in harmony with nature,” Saman said.

Leila replied, “Exactly. Every system—from energy and water to transportation and waste—operates in alignment with both citizen well-being and the environment.”

Reduced noise and air pollution, safe mobility paths, and accessible green spaces significantly enhance physical and mental health. Real-time algorithms preserve environmental quality and alleviate urban stress by continuously maintaining balance.

Modern cities, burdened by traffic, pollution, and systemic fragmentation, typically provide a far lower level of psychological well-being.

Chapter 3: Concluding Inference

Niyaee City, as observed through the searching eyes of Ava and Ramin and the curious gazes of Leila and Saman in the previous two chapters, is not merely a modern example of interaction between humanity and technology. Rather, it stands as a vivid manifestation of a new urban philosophy—one in which urban life, nature, and artificial intelligence are woven together as a single, living, and integrated whole.

As the morning mist settled upon the honeycomb-like walls of the city, refracting light into a thousand hues, one could sense not only visual beauty but also the intrinsic harmony of its systems—a harmony made possible by Bee Colony Optimization (BCO) algorithms and an extensive network of sensors.

1. The City as a Living Ecosystem

Niyaee City cannot be understood as a mere aggregation of buildings and streets; it is a dynamic ecosystem in which every unit—whether a sensor, a plant, or a citizen—plays a vital role in maintaining overall balance. Bee colony algorithms manage not only the flows of energy, water, and waste, but also enable citizens themselves to become integral

components of the system. Every small human decision or environmental fluctuation, much like the movement of a single bee within a colony, generates feedback that optimizes the entire network.

Unlike traditional cities, Niyaee City exhibits no sudden disruptions or wasteful dissipation of resources. Energy consumption, water flow, building ventilation, and the movement of light electric vehicles are all predicted and regulated in a coordinated, rhythmic choreography. This harmony enhances not only system efficiency but also the psychological comfort and well-being of citizens. In essence, Niyaee City is a living organism, each of its cells resonating in unison with artificial intelligence and nature.

2. Artificial Intelligence and Bee Colony Algorithms: A Bridge Between Technology and Nature

BCO algorithms, inspired by the collective behavior of bees, extend far beyond conventional resource management. They synchronize complex, real-time decisions using vast urban datasets. By observing and analyzing citizen behavior, these algorithms optimize resource allocation, manage energy consumption

peaks, and even anticipate environmental crises.

Functioning as a form of collective intelligence, these algorithms enable the city to operate in a self-regulating manner. Unlike traditional energy networks—which are reactive and vulnerable—Niyaee City remains resilient under sudden fluctuations and pressures. Like bees within a colony, the algorithms compensate for shortages, provide immediate feedback, and prevent the waste of energy and resources. This performance is not merely technological but is deeply aligned with ecological principles and sustainable urban living.

3. Environmental Education and the Role of Citizens

One of the most distinctive features of Niyaee City is its practical, real-time environmental education for citizens. From early childhood, children engage with environmental games and intelligent feedback systems that teach them how to optimize the use of energy, water, and food. Algorithms analyze and gently correct behaviors, aligning individual actions with the broader urban ecosystem.

This approach stands in stark contrast to conventional

cities, where environmental education is often limited and its influence on daily life minimal. Through hands-on learning and intelligent feedback, Niyaee City transforms citizens into active, responsible participants within the ecosystem.

4. Vertical Gardens, Microclimates, and Resource Reduction

Greenhouses and vertical gardens serve not only aesthetic purposes but also perform critical environmental functions. These systems regulate the city's microclimate, improve air quality, and reduce energy consumption for ventilation and lighting. Intelligent algorithms use real-time sensor data to adjust irrigation, airflow, and illumination pathways, delivering rapid and optimal responses even during environmental crises.

This symbiotic interaction between technology and nature reduces water and energy consumption by 30–40 percent compared to traditional cities, demonstrating that intelligent coexistence with nature is not only possible but essential.

5. Waste Management and the Closed Energy Cycle

Another hallmark of Niyaee City is its intelligent waste

management system. Each unit of waste carries a smart identifier, allowing sensors to detect its type, volume, and chemical composition. Algorithms then determine the optimal pathway—recycling, energy generation, or compost production. Immediate feedback provided to citizens helps refine environmentally responsible behavior.

This system creates a closed-loop cycle that reduces greenhouse gas emissions and ensures optimal resource utilization. Here again, bee colony algorithms mirror nature itself: each cell fulfills its role, and the entire system achieves exceptional efficiency.

6. The Sustainable City: Inspired by Nature, Realizing Human Well-Being

Niyaee City exemplifies a model in which human well-being and environmental preservation are achieved simultaneously. Reduced air and noise pollution, safe pathways, and widespread access to green spaces enhance both physical and mental health. Energy and water consumption are optimized, peak loads are managed, and environmental crises are addressed without disrupting daily life.

The ultimate message is that urban sustainability is not

merely a technological objective, but a deeply human and ethical one. Artificial intelligence, bee colony algorithms, and sensor networks are tools toward this end—but the true spirit of a sustainable city lies in the harmony between humanity, technology, and nature.

7. Golden Lessons

1. **Macro–Micro Harmony:** A sustainable city functions coherently at both systemic and granular levels.
2. **Real-Time Feedback and Self-Regulation:** AI-driven feedback minimizes waste and enhances efficiency.
3. **Citizen Education and Responsibility:** Informed and engaged citizens are the cornerstone of sustainability.
4. **Learning from Nature:** Bee colony algorithms reveal nature as the ultimate teacher of intelligent system design.
5. **Human Well-Being and the Environment:** Environmental protection and human welfare must be pursued together, with technology serving as a means—not an end.

Ultimately, Niyaee City is not a dream, but a clear blueprint for a future in which urban life is harmonized with intelligence, precision, and beauty. Just as the morning mist settles upon the city and refracts light into countless colors, so too can the harmony between humanity, nature, and technology transform urban life into an experience that is elegant, dynamic, and sustainable. Bee colony algorithms, vertical gardens, and sensor networks are not merely technical instruments; they are symbols of a philosophy of life—one in which every movement, every decision, and every cell is part of a living, intelligent whole.

The Colony Notebook – Final Reflective Questions (Two Sections)

Section One: 40 Questions for Reflection and Analysis

1. In the structure of **Niyaee City**, which elements cause the city to function like a "living organism," and to what extent is this metaphor scientifically defensible?
2. In the honeycomb-inspired architecture of Niyaee City, what relationship is established between aesthetics, energy efficiency, and sustainability?
3. If Niyaee City were operated without bee colony algorithms, which urban systems would experience the greatest decline in efficiency?
4. What similarities exist between resource management in Niyaee City and material cycles in natural ecosystems?
5. Can Niyaee City be considered a "cyber–biological ecosystem"? Why or why not?
6. How do bee colony algorithms in Niyaee City

simulate “collective decision-making” at an urban scale?

7. What role do BCO algorithms play in predicting environmental crises, and what limitations might they have?

8. If a key element of the city’s sensor network fails, can colony algorithms self-repair in a manner similar to natural systems?

9. How does the “collective intelligence” of bee-inspired algorithms differ from classical machine learning models?

10. Can BCO algorithms be regarded as an ethical model for sustainable urban governance?

11. What role do vertical gardens in Niyaee City play in mitigating the urban heat island effect?

12. Why can the closed-loop energy cycle of Niyaee City be considered a “radical sustainability model”?

13. Can it be argued that Niyaee City has achieved the concept of “zero waste”? Why or why not?

14. To what extent has the selection of native plants

in microclimate design contributed to reducing energy consumption?

15. If annual rainfall were to decrease by 40 percent, how should the city's intelligent systems be redesigned?

16. What kind of "learning ecological behavior" emerges from citizens' interaction with Niyaee City?

17. Does real-time AI feedback risk creating a form of behavioral dependency among citizens?

18. To what degree is the psychological calm of Niyaee City's citizens a result of physical design versus intelligent management?

19. What existential implications arise from children's interactions with artificial intelligence systems?

20. Does living in a highly intelligent city reduce freedom of choice, or does it enhance it?

21. What vulnerabilities might a BCO-based intelligent network face in the event of cyberattacks?

22. What distinguishes the data-driven waste management of Niyaee City from classical urban management models?

23. To what extent can biosensors produce errors, and how do bee colony algorithms compensate for them?

24. Can algorithms develop data biases toward citizen behavior?

25. If the city's population were to increase fivefold, how should colony algorithms be redesigned?

26. What metaphorical function do Leila and Saman serve in the narrative?

27. Why is the architecture of Niyaee City presented as a "dialogue between humanity and nature"?

28. How is the contrast between modern cities and Niyaee City emphasized through literary description?

29. Why does the author repeatedly emphasize the "breathing of the city," and what does this metaphor signify?

30. Are "light," "air," and "plants" primarily

descriptive or conceptual elements in the narrative? Analyze.

31. In Niyaee City, does ethical responsibility lie more with humans or with algorithms?

32. Can algorithms replace human ethical judgment in environmental protection?

33. What vision of the future relationship between humanity and nature does Niyaee City present?

34. To what extent can artificial intelligence systems be entrusted with “environmental trust”?

35. Should a sustainable city be designed “without human emotions,” or precisely the opposite?

36. If a modern city were to implement only 30 percent of Niyaee City’s principles, which components would have the greatest impact?

37. Why are today’s cities unable to easily adopt the Niyaee City model?

38. Can Niyaee City be described as a “point of equilibrium” between technology and nature? Why?

39. Which technologies used in Niyaee City are economically feasible for real-world cities today?

40. To what extent is Niyaee City a utopian vision, and to what extent is it an achievable model?

Section Two: 40 Practical Exercises

1. Build a simple BCO model with 20 agents that optimizes the energy consumption of a building over 24 hours. Plot the results.

2. Assuming 500 environmental sensors, design a scenario in which the city responds automatically to a temperature change within 10 seconds.

3. Design a crisis prediction system (e.g., water scarcity or CO_2 increase) using hypothetical data.

4. Write an algorithm that adjusts street lighting intensity based on human presence, animals, and weather conditions.

5. Design a bee-hive-inspired collective decision-making model to determine which street should

receive the most solar energy.

6. Design a small urban block whose form is based on the structure of a beehive.

7. For three building types (residential, administrative, public), design natural ventilation patterns inspired by bee nests.

8. Reconstruct wind flow patterns in Niyaee City and demonstrate how organic architecture guides airflow.

9. Design a vertical garden with 20 plant species and explain each plant's role in the microclimate.

10. Design a system showing how waste from a single building can be converted into electrical energy.

11. Generate a dataset of 30 hypothetical values (temperature, humidity, wind speed, light) and analyze how the city automatically balances them.

12. Design an air pollution scenario and show how the city restores equilibrium within five minutes.

13. Plot solar energy fluctuations throughout the day

and analyze how the city redistributes energy loads.

14. Model a family's water consumption and explain how Niyaee City's intelligent purification system recycles 70 percent of it.

15. Design a diurnal temperature pattern for a one-square-kilometer area and analyze the role of vegetation and wet surfaces.

16. Write a two-page essay: if your city were to become similar to Niyaee City within ten years, what changes should begin now?

17. Write a narrative scene in which an environmental crisis occurs and the city manages it autonomously.

18. Write a descriptive page of Niyaee City's architecture without using any technological terms.

19. Write an analytical page on the disasters that would result from the absence of such a city in the future.

20. Write a two-page article on the concept of the

"breathing city": its meaning, metaphor, and scientific function.

21. Illustrate the complete energy cycle of Niyaee City (from solar source to end user).

22. Build a solar energy model for a cloudy day and analyze how the city compensates for shortages.

23. Design a "zero-waste network" map: separation, processing, and energy conversion.

24. Write a scenario in which 30 percent of solar panels fail and the city must reorganize itself.

25. Design and analyze the stages of neighborhood-scale biogas production.

26. Design three behavioral patterns through which the city adapts to citizen behavior (e.g., lighting, ambient music, pedestrian routes).

27. Design an environmental education system for children: scenario, game mechanics, data, and experience.

28. Design the city's response when 20 children suddenly enter a green space during playtime.

29. Write a dialogue between a child and the city's AI system about the importance of water.

30. Design a communication model showing how the city conveys messages through non-verbal language (light, wind, music, color).

31. Design a scenario for a small fire in a vertical garden and force Niyaee City to respond naturally.

32. Design a ten-step scenario for managing energy shortages on a cloudy morning.

33. Model a flood warning system using ten sensors.

34. Design a hypothetical cyberattack on the energy network and analyze the AI's defensive response.

35. Design a model for emergency water distribution during severe shortages.

36. Write one page on the "boundary between human decision-making and algorithmic authority" in Niyaee City.

37. Propose a strategy to increase citizens' trust in artificial intelligence systems.

38. Write a scenario in which the city must choose between human comfort and environmental preservation.

39. Analyze how the city's behavior would change if Niyaee City possessed "emotions."

40. Design a philosophical model explaining why Niyaee City resembles a living organism.